

AI Resume Builder - Frontend

A modern, interactive web application built with Next.js that guides users through creating professional resumes using an AI-powered (or rule-based) chatbot interface.

■ Features

- **Conversational UI:** A sleek, step-by-step chatbot interface to collect user information.
- **Progress Tracking:** Real-time progress bar to show how much of the resume is completed.
- **Dynamic Preview:** See your information being formatted as you provide it.
- **Instant PDF Download:** One-click generation and download of professional PDF resumes.
- **Responsive Design:** Optimized for both desktop and mobile devices.
- **Smooth Animations:** Powered by Framer Motion for a premium user experience.

■■ Tech Stack

- **Framework:** [Next.js 15](https://nextjs.org/) (App Router)
- **Styling:** [Tailwind CSS 4](https://tailwindcss.com/)
- **UI Components:** [Radix UI](https://www.radix-ui.com/) & [Lucide Icons](https://lucide.dev/)
- **Animations:** [Framer Motion](https://www.framer.com/motion/)
- **State Management:** [Zustand](https://github.com/pmndrs/zustand)
- **API Client:** [Axios](https://axios-http.com/)
- **PDF Rendering:** [@react-pdf/renderer](https://react-pdf.org/)

■ Prerequisites

- Node.js 18+
- npm / yarn / pnpm

■■ Setup & Installation

1. Clone the repository:

```
git clone <repository-url><br/> cd AI_RESUME_GENERATOR_FRONTEND
```

2. Install dependencies:

```
npm install
```

3. Configure Environment Variables:

Create a `.env.local` file in the root directory:

```
NEXT_PUBLIC_API_URL=http://localhost:8000
```

4. Run the development server:

```
npm run dev
```

5. Build for production:

```
npm run build<br/> npm run start
```

■■ Project Structure

```
src/<br/>■■■■ app/ # Next.js App Router (pages and layouts)<br/>■ ■■■■ builder/ # Main resume builder interface<br/>■ ■■■■ resume/ # Resume view/management<br/>■■■■ components/ # UI components<br/>■ ■■■■ chatbot/ # Chat interface components<br/>■ ■■■■ pdf/ # PDF generation components<br/>■ ■■■■ preview/ # Real-time resume preview<br/>■ ■■■■ ui/ # Base shadcn/ui components<br/>■■■■ services/ # API communication (Axios)<br/>■■■■ store/ # Global state (Zustand)<br/>■■■■ types/ # TypeScript definitions<br/>■■■■ utils/ # Helper functions
```

■ Architecture

The application follows a modern decoupled architecture with a centralized state management system.

```
graph TD<br/> subgraph "Frontend (Next.js)"<br/> UI[User Interface - React Components]<br/> CS[Chat Store - Zustand]<br/> RS[Resume Store - Zustand]<br/> API[API Client - Axios]<br/> PDF[PDF Renderer - @react-pdf]<br/> end<br/><br/> subgraph "Backend (FastAPI)"<br/> BE[Resume Generation API]<br/> end<br/><br/> UI <--> CS<br/> UI <--> RS<br/> CS --> RS<br/> RS --> PDF<br/> RS --> API<br/> API <--> BE
```

■ Application Flow

The following flowchart illustrates the user journey and data flow within the application.

![Frontend Workflow](./flowchart.png)

Sequential Flow (Mermaid)

```
sequenceDiagram<br/> participant User<br/> participant Chat as Chat Interface<br/> participant Store as Zustand Store<br/> participant Preview as Live Preview<br/> participant API as Backend API<br/> participant PDF as PDF Generator<br/><br/> User->>Chat: Provides Personal Info<br/> Chat->>Store: Updates Resume Data<br/> Store->>Preview: Triggers Re-render<br/> Preview-->>User: Shows Live Change<br/><br/> Note over User, Preview: Repeat for Experience, Skills, etc.<br/><br/> User->>Chat: Completes Final Step<br/> Chat->>Store: Finalize Data<br/> Store->>API: Send Data for Backend Processing<br/> API-->>Store: Returns Processed Data/Blob<br/> Store->>PDF: Generate Downloadable PDF<br/> PDF-->>User: Triggers File Download
```

■ Generated PDF Documentation

If you are unable to download the PDF files directly from the links provided by the assistant, you can find them in your local project directory:

- **Frontend PDF:**

`c:\Users\samee\OneDrive\Desktop\AI\_RESUME\_GENERATOR\_FRONTEND\README.pdf`

- **Backend PDF:**

`c:\Users\samee\OneDrive\Desktop\AI\_RESUME\_GENERATOR\_BACKEND\README.pdf`

To open them, simply navigate to the folders in your File Explorer and double-click the files.

■ Integration

This frontend is designed to work seamlessly with the [AI Resume Builder Backend](https://github.com/Yashwant-Rangrej/AI_RESUME_GENERATOR_BACKEND). Ensure the backend is running and the NEXT_PUBLIC_API_URL is correctly configured.

■■■ Developed By

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